TIMOTHY HERRERA

Robotics Operations Shift Lead | Robotics Fleet Operations | Data Collection & Quality | Team Leadership | Floor Safety
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PROFESSIONAL SUMMARY

Robotics operations leader who served as Operations Coordinator & Shift Supervisor at The Bot Company, a San Francisco AI robotics company building a general-purpose home robot. Ran day-to-day floor execution for a robot data collection team: scheduling operators, keeping stations and workflows covered, owning data quality, and escalating hardware and data defects to research, hardware, and engineering teams. Promoted from Research & Development Operator within seven months; onboarded and trained 70+ new hires and authored the SOPs, troubleshooting guides, and safety training the floor runs on. Maintained a 150-robot humanoid fleet with firmware kept current across sites. Hands-on with robot hardware troubleshooting, robot recovery and scene resets, and Foxglove/SSH diagnostics to keep collection running. Built and deployed a full-stack robot fleet operations dashboard (Python/FastAPI, React/TypeScript) with predictive maintenance alerts, shift handoff reports, and capacity planning, plus a 3D robot arm simulator with inverse kinematics, sensor-driven pick-and-place, and two-arm task coordination. Backed by a decade-plus career in data operations and QA across fintech, healthcare, and airline accounts.

CORE SKILLS

- **AI Agentic Engineering:** Shipping full-stack apps with AI coding agents (Claude Code): scoping agent tasks, writing specs, reviewing and testing agent-written code, verifying live before deploying
- **Floor Operations:** Daily shift execution; operator scheduling and work assignment; station, robot, and staffing readiness; maintaining coverage and workflow continuity through disruptions
- **Team Leadership:** Supervising, training, and coaching operators; onboarding 70+ new hires; enforcing work standards and quality expectations; relaying operator concerns to management
- **Throughput & Data Quality:** Throughput and data-quality KPI tracking; validating robot training data against standards; identifying deviations, errors, and edge cases; defect logging and escalation
- **Cross-Functional Communication:** Primary floor contact for ML, robotics, AI, and hardware engineering teams; clear reporting of technical blockers, hardware failures, and process bottlenecks
- **Safety & Continuous Improvement:** Enforcing safe robot-operation and lab protocols; routine safety checks; incident logging, investigation, and escalation; SOP authoring and workflow standardization
- **Robotics Tools & Troubleshooting:** Foxglove telemetry dashboards; SSH remote control and diagnostics; Linux command line; diagnosing sensors, actuators, connectors, and mechanical assemblies; teleoperation (VR, haptic/handheld controllers, motion tracking); robotic manipulation (pick-and-place); robot recovery and scene resets; robot kinematics (forward/inverse: analytic, Jacobian, CCD), motion planning, and multi-arm collision checking
- **Robotics Fleet Operations:** Maintaining a 150-unit humanoid robot fleet; keeping firmware up to date and deployed across sites; ensuring robots are clean and deploy-ready; robot and equipment logistics
- **Fleet Analytics:** Ops KPI design (QA pass rate, goal attainment, availability, MTBF/MTTR, utilization, cost per usable episode, shift fill rate, incident rate); predictive maintenance scoring; capacity and downtime planning
- **Data & Technical:** Advanced Excel (VLOOKUP, pivot tables); Python (FastAPI, pandas, NumPy, pytest) and SQL (SQLite/PostgreSQL) for APIs, KPI queries, and validation scripts; React + TypeScript dashboards; Docker; cloud deployment (Render); CSV/JSON; Git/GitHub; MERN stack foundation

EXPERIENCE

Operations Coordinator & Shift Supervisor | Sep 2025 – Sep 2026
The Bot Company — San Francisco, CA (AI robotics, general-purpose home robot)

- Promoted from Research & Development Operator after seven months for strong performance on data quality and team coordination.
- Led daily on-site floor operations as Shift Supervisor for the data collection team: scheduled operators, assigned work, and kept coverage and workflow continuity through disruptions and staffing gaps.
- Owned data quality for the team's output, validating robot training and evaluation data for accuracy and consistency before it entered the training pipeline, and reviewing model outputs to flag errors, edge cases, and coverage gaps.
- Tracked throughput and data-quality KPIs; logged defects and escalated data, robot, and hardware issues to robotics, AI, and hardware engineering teams.
- Served as a primary floor point of contact for ML engineers and cross-functional teams, communicating status, technical blockers, and process bottlenecks.
- Trained and onboarded 70+ new Research & Development Operators on robot operation, data collection procedures, and quality standards; provided ongoing coaching on work standards.
- Enforced safe robot-operation and lab protocols on the floor, ran routine safety checks, and logged, investigated, and escalated incidents to keep the team compliant during data collection.
- Authored SOPs, troubleshooting guides, and safety training for robotics data collection and model learning tasks, standardizing workflows across the team.
- Maintained a fleet of 150 humanoid robots, keeping firmware up to date and deployed across sites and ensuring every robot was clean and deploy-ready.
- Minimized downtime by supporting robot hardware troubleshooting and Foxglove/SSH-based diagnostics; used Foxglove daily to monitor and review robot data and telemetry.

- Handled logistics and owned office and facilities coordination for a 100+ person site, including vendor management, supply and equipment logistics, and workspace setup.

Research & Development Operator | Feb 2025 – Aug 2025
The Bot Company — San Francisco, CA
- Operated and monitored robots via teleoperation during data collection sessions capturing real-world household task data.
- Performed robotic manipulation tasks, including pick-and-place, and handled robot recovery and scene resets to keep data collection sessions running safely.
- Collected and validated training and evaluation data for accuracy, consistency, and quality before it entered the training pipeline.
- Diagnosed and fixed robot hardware issues (sensors, actuators, connectors, mechanical assemblies) to keep sessions running.
- Used Foxglove to visualize telemetry and sensor data, and SSH to command robots remotely for diagnostics and testing.

PROJECTS

Robot Fleet Ops Metrics — Operations Dashboard | Nov 2026
Live demo: robot-fleet-ops.onrender.com · **Code:** github.com/timHfleet/robot-fleet-ops
- Built and deployed a 12-page operations dashboard for a robot data-collection fleet tracking the KPIs I ran a shift on: QA pass rate, goal attainment, availability, MTBF/MTTR, utilization, cost per usable episode, and incident rate.
- Migrated it off a low-code (Retool) prototype, porting 20 backend functions from PostgreSQL to a Python/FastAPI API and rebuilding the UI and data layer in React/TypeScript.
- Designed an explainable predictive maintenance score (0–100) comparing each robot's last 7 days with its own 4-week baseline (failure rate, repair time, availability), with a recommended action for each robot.
- Added a shift handoff report exportable as Slack-ready text or PDF, and a capacity planner showing how many robots can go down, and for how long, before output drops below 90% of the weekly target.
- Added a root-cause view splitting each KPI change by robot, shift, operator, and reject reason, plus a management report.

Robot Arm Simulator — Kinematics, Motion Planning & Two-Arm Tasks | Nov 2026
Live demo: robot-2arms-sim.onrender.com · **Code:** github.com/timHfleet/robot-simulation
- Built and deployed a 3D simulator (Python/FastAPI + NumPy, React/TypeScript + three.js) for a 5-joint arm, solving inverse kinematics three ways: analytic, Jacobian (damped least squares), and CCD.
- Planned pick-and-place like real robots run it (camera scan, close-up correction, slow final approach, grip-force sensing, recovery from slips and drops) and household tasks from real data-collection work: loading a dishwasher and folding a towel and T-shirt.
- Coordinated two arms with a frame-by-frame collision-checking scheduler, loading the dishwasher 1.7× faster than one arm, plus lockstep two-arm cloth folding; 68 automated tests.

Data Operations Analyst | Dec 2018 – Jan 2020
ECE Consulting Group — Account: Robinhood (fintech trading platform)
- Processed and maintained customer account records for a high-growth fintech platform; performed quality checks on account and transaction records against source data.

Customer Care Specialist | Mar 2016 – May 2017
Concentrix — Account: United Airlines (MileagePlus)
- Managed reservations, ticketing, and account updates; conducted QA reviews of reservation and account data for accuracy and policy compliance.

Senior Process Executive | Feb 2015 – Oct 2015
Cognizant — Account: Computershare Limited
- Validated shareholder records before processing and resolved data discrepancies directly with account holders.

Insurance Verification Specialist | Jun 2014 – Jan 2015
Arriva Medical, LLC
- Verified insurance coverage, processed supply orders, and posted client payments as Accounts Receivable and Cash Posting Agent.

Medical Chart Specialist | Jan 2014 – Jun 2014
Altegra Health Inc.
- Imported and audited client roster records against source data, flagging inconsistencies to determine reimbursement status.

Senior Business Process Associate | May 2012 – Jan 2014
UST Global — Account: Title Source, LLC
- Processed and validated mortgage profile data for accuracy and completeness ahead of tax reporting and compliance.

EDUCATION

Full-Stack Web Development (MERN Stack) | Apr 2022 – Apr 2023
Zuitt Coding Bootcamp, Batch 176 — Philippines
- Built and deployed a MERN-stack booking application (React front end, Express/Node REST API, MongoDB Atlas); tested APIs with Postman.

Bachelor of Science in Information Technology
Northeastern College, Isabela, Philippines (2010–2011) | Isabela State University, Isabela, Philippines (2009–2010)